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An Improved Bus Travel Time Prediction using Multi-Model Ensemble Approach for Heterogeneous Traffic Conditions

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***Abstract*— An accurate and reliable arrival time prediction of buses to the next bus stops is a valuable tool for both passengers and operators. Existing studies have some limitations in bus travel time prediction. They focus little on three aspects such as heterogeneous traffic flow conditions, dwell time prediction and interpretation of explanatory variables. Consequently, we break down the prediction problem into sub-models for running time and dwell time prediction and incorporate a feature engineering framework that generates features related to the running bus, the prediction day, and immediate and historical time variations to capture heterogeneous traffic conditions. We propose a multi-model stacked generalisation ensemble model by leveraging the advantages of best-performing models in homogeneous conditions such as Extreme Gradient Boosting (XGBoost) and convolutional long short-term memory (ConvLSTM) models. It outperformed the state-of-the-art models by 11% in mean absolute error (MAE) on average. It can predict extreme conditions in bus journeys more accurately. When interpreting the feature importance, we found insights like driver behaviour and preceding travel time influence the prediction model, which paves the way for strategic management by authorities.**

I. Introduction

Bus travel time prediction is one of the paramount components of intelligent transportation systems (ITS) and has gained increasing importance over the past few decades. Accurately predicting the bus travel time is crucial as it is affected by factors like rush hours, weather, and passenger demand. Moreover, travel time prediction has become more rigorous in heterogeneous traffic conditions [1]. It is also known as mixed-traffic conditions, which is mainly observed in developing Asian countries, indicating that the roadway is simultaneously shared by various vehicle classes, such as buses, trucks, cars, three-wheelers, and motorcycles. It causes a lack of lane discipline, congestion potential, and absence of bus priority measures leading buses to mix with the general traffic, and consequently, they compound the prediction problem crucial. At the same time, from a study conducted in Sri Lanka [2], it is observed that the bus dwell times show a significant spatial and temporal variation, hence, travel time varies. Therefore, there is a need for a precise travel time prediction model for buses in developing countries with heterogeneous traffic conditions.

The predicted arrival times to the bus stops along the journey contribute much to developing Advanced Traveler Information Systems (ATIS). The ATIS is beneficial to passengers, drivers, and transit operators/authorities. The reliable bus service with the arrival time information reduces the anxiety among the passengers about the bus arrival and supports them in trip planning and decision making. In addition, due to the uncertainty and dynamicity in heterogeneous traffic, drivers can make strategies to adhere to the schedule. Transit authorities can optimize the schedule in real-time by observing the delays and variations in the heterogeneous traffic and can command to reduce bus bunching [3].

A typical bus journey with a bus, when reaching a stop i, is depicted in Fig 1. Technically, an ATIS needs to predict

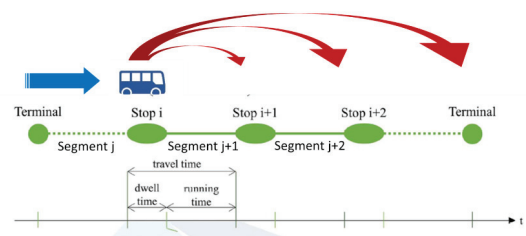

Fig. 1. An overview of the problem of predicting the arrival times to the next bus stops incorporating running time at segment and dwell time at stop

the arrival times to the successive downstream bus stops $(i+1, i+2, \ldots)$ given the history of the trip so far up to stop i and other impacting conditions in real-time. Travel time taken by the bus to reach the next stop $i+1$ is called the 'running time' (RT) in the segment $j+1$, and the time spends at stop i, in waiting to drop off and pick up passengers is known as the 'dwell time' (DT) at stop i. With that, when the bus reaches the bus stop i, as given in equation (1), the arrival time (AT) to the next stop $i+1$ is calculated.

$$AT_{i+1} = DT_i + RT_{j+1} \tag{1}$$

The dwell time distribution is generally uniform in homogeneous traffic conditions [4]. However, in heterogeneous conditions, the variations of dwell times at different bus stops are significantly large, than the average. In addition, this study is designed to predict the arrival time to the next stop as a single-step prediction, in contrast to the multi-step prediction of all the downstream bus stops, at once.

A wide variety of approaches are used in predicting travel time in homogeneous conditions [5]. Existing studies [6], [7] have predominantly developed models by dividing the routes into segments by either bus stops [8] or inter-sections [6] and then, predicting the travel times. However, some studies [9], [10] implicitly considered the effects of dwell times as the input features using the data on passenger demand.

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Manuscript 881 submitted to 2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC). Received May 26, 2023.

Further, a few more studies [11], [12] have paid attention to developing a dwell time prediction model separately in conjunction with the travel time model. Still, they adopted more simplistic models. Correspondingly, Wu et al., [13] developed a model merging the predictions from separate segment time and dwell time models. However, applying those varieties of models with data from heterogeneous traffic lagged significantly in accuracy [1]. Besides, prior approaches failed to well explain the contributing factors to the bus arrival time prediction, which can improve the reliability of prediction in heterogeneous traffic.

In summary, few attempts to predict travel times in heterogeneous conditions could not significantly perform well. They lack segregation of dwell time and running time to understand the implications of dynamicity in heterogeneous traffic and they lack explainability for stakeholders to understand and make meaningful decisions. Hence, passengers lose trust in this system and authorities could not develop strategies to improve the reliability of the bus service.

In light of the above, we propose a multi-model ensemble approach by integrating independent models for predicting running times and dwell times. The multi-model allows us to analyze dwell times in detail by capturing the variations in heterogeneous conditions. From the recent studies, a deep learning model, convolutional long short-term memory (Conv-LSTM) [14] performed well, while extreme gradient boosting (XGBoost) has higher accuracy in the sets of tree-based ensemble models [8]. Combining the predictions of multiple models makes achieving higher accuracy than any individual model can achieve. Therefore, we used stacked generalization [15] as an effective ensemble method, especially when the models being combined are diverse and complementary. Further, we develop a feature engineering framework that generates features that depict the travel time variability in heterogeneous conditions. Finally, we provide insights to decision makers with feature important analysis. All code used in the model development is available online[1].

The rest of this paper is structured in the following order: Section II reviews the related works and Section III provides a detailed explanation of the proposed methodology with feature engineering, and the model development. Later, section IV explains the experimental design. Next, the results obtained from the ensemble model are compared with the base models in Section V. Finally, the conclusions and future research directions are outlined in Section VI.

II. Related Works

The bus travel time prediction problem is not widely addressed in heterogeneous traffic conditions [1]. The models developed in homogeneous traffic underperformed in heterogeneous traffic conditions. A few researchers from India [16]–[19], developed some techniques for predicting bus travel time in heterogeneous conditions. Initially, the authors of [16] adopted a simple Kalman filter (KF) technique in this case. Then, another researcher [17] introduced a time-space discretization approach that considers both temporal and spatial variations observed in mixed traffic. Later, a spatial KF kernel was developed [19] to capture real-time uncertain variation in traffic state and update the arrival times. However, this model exploits only the temporal correlations of the previous bus. Hence, advanced models need to be explored in heterogeneous conditions to capture variations.

Most of the previous studies dealt with homogeneous traffic conditions. Earlier approaches were based on historical averaging [20] and statistical models like seasonal auto-regressive [18]. Later, machine learning (ML) models like support vector machine (SVM) [21], k-nearest neighbour (kNN) [22] and artificial neural networks (ANN) [7] were utilized. However, with the recent advancement in deep learning, researchers showed that travel time prediction can be further improved by adopting deep learning architectures such as LSTM [9], [23] which can learn temporal correlations. Furthermore, a hybrid model as a combination of convolutions and LSTM layers into a single structure, ConvLSTM was applied [11] and it outperformed other deep learning models [24]. Here, the convolution layer could learn spatial dependencies between segments. In parallel, ensemble learning became popular in travel time prediction. Bagging ensembles (eg: Random forest (RF) [25]) and gradient boosting ensembles (eg: XGBoost [8]) were used. XGBoost outperformed other ML models, and it improved accuracy while reducing the occurrence of over-fitting. Besides, there is no standard data set used by these researchers, which reflects no direct comparison. However, the above models were developed for a homogeneous traffic stream. A study [1] conducted in India, compared modern ML algorithms and showed that their results are lagging behind those of developed countries with homogeneous traffic.

On the other hand, the multi-model approach for travel time prediction is not been extensively explored. The authors of [11] introduced it. They used a simplistic dynamic exponential smoothing (ES) for dwell time prediction since they could not capture latent travel demand. Then, another scholar [12] adopted a simple model with a weighted KNN. Recently, deep learning (ConvLSTM) models were tried for both dwell time and running time [13]. However, it failed with low accuracy in heterogeneous traffic. Yet again, the multi-model approach is not explored in heterogeneous conditions. In addition, the driver behaviour of buses will be disparate in the heterogeneous, disordered traffic [1]. However, it was not incorporated into any model development. Therefore, more attention needs to be paid to generating real-time features and developing models that can learn the non-linear correlations for bus arrival time prediction in heterogeneous conditions.

III. Methodology

In this section, we present our proposed multi-model ensemble approach to improve the accuracy of the bus arrival time prediction. The high-level architecture of the system is shown in Fig. 2. The step-by-step detailed explanation is elaborated in the following sub-sections.

[1] https://github.com/aaivu/MME4BAT

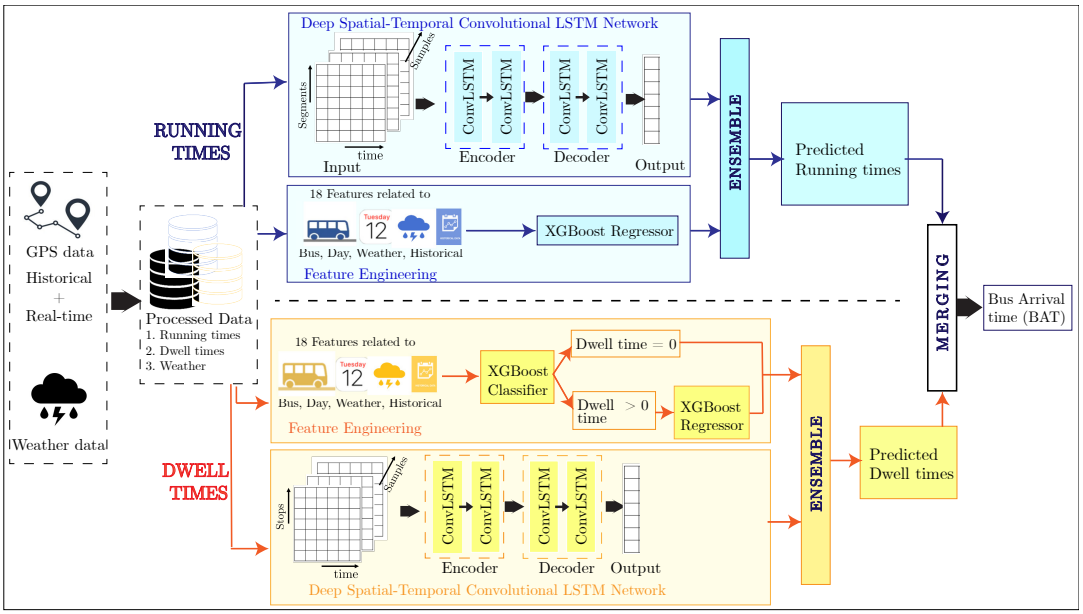

Fig. 2. High-level architecture of the proposed multi-model ensemble approach in predicting bus arrival times segregating the individual predictions of bus running time and dwell time which were obtained by the ensemble of ConvLSTM and XGBoost regressor models. It shows the process starting from data processing, feature engineering, encoder-decoder arrangement of ConvLSTM cells, predictions and merging to result in arrival time to next stop

A. Feature engineering

We have generated features to capture the spatial and temporal dependencies of both running and dwell times, which, in a sense, can replicate heterogeneous variability. Moreover, they are real-time features depicting the influence of the 'target bus', the 'predicting day' conditions, and the immediate historical times accommodated in the target bus stop or segment, that is, travel times from the previous weeks at a similar day and time slot. The description of the 18 spatial-temporal generated features is listed in Table I. We have divided every hour of the day into 15-minute resolutions. We have used some novel ideologies in feature generation in addition to the existing studies for freeway travel time prediction [26]. For example, dwell time at the previous bus stop impacts the running time in the current segment (inter-model feature). In addition, driver behaviour influences variations in travel time. We included $'device_id'$ as a feature that reflects the target bus in the prediction. Furthermore, the running/dwell times of the predicting bus in its last three segments/stops can indicate the behavioural pattern. Similar and symmetrical 18 features are generated for the dwell time model too. Tree-based models are good at handling more features, even correlated ones. At the same time, encoder-decoder architecture was adopted for deep feature extraction in the ConvLSTM deep learning model.

TABLE I

THE INPUT FEATURES AND DESCRIPTION FOR RUNNING TIME MODEL

Category	Feature	Description
General	*DoW*	Day of the week
	ToD	Time of Day (every hour divided to 15-minute horizons)
	Holiday	Holiday or Sunday
	Saturday	Saturday of a week
	device_id	Unique identification for buses
	segment_id	Unique identification for segments
	length	Length of the segment
Weather	*precip*	Precipitation at that time horizon
	temp	Temperature at that time horizon
Target Bus	$RT(j-1)$	Running time of the bus at the previous upstream segment
	$RT(j-2)$	Running time of the bus at the second immediate segment
	$RT(j-3)$	Running time of the bus at the third immediate segment
Predicting day	$RT(t-1)$	Average running time at the target segment 15 min before
	$RT(t-2)$	Average running time at the target segment 30 min before
Historical	$RT(w-1)$	Running time at the segment in similar time, 1 week before
	$RT(w-2)$	Running time at the segment in similar time, 2 week before
	$RT(w-3)$	Running time at the segment in similar time, 3 week before
Previous stop	$DT(n-1)$	Dwell time at the previous bus stop

B. Model development

In this study, we exploited two recent state-of-the-art models in travel time prediction used in homogeneous traffic, ConvLSTM, and XGBoost. The ConvLSTM model was introduced for weather forecasting [27], which is a type of RNN, with a novel combination of convolution and LSTM layers. The convolution layer learns close neighbours in the input tensor. An LSTM network is a chain of repeating modules called *cells* (C_t) and contains gates (i_t: input gate, O_t: Output gate & f_t: forgetting gate) that selectively remember/forgot information. As shown in Fig 3, ConvLSTM is an extension of LSTM which has convolution layers in both the input-to-state and state-to-state transitions. Then

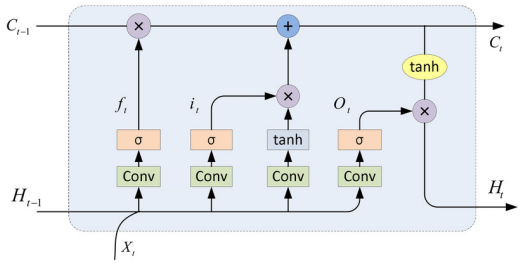

Fig. 3. The structure of the ConvLSTM cell where Convolutional layers (shown in the green colour box) are inserted into LSTM structure

four ConvLSTM cells are stacked to make encoder-decoder architecture. Finally, the fully connected layer is replaced by the time-distributed layer to provide multi-outputs. In addition, *drop out* layers are added to avoid over-fitting and *Batch normalisation* to improve performance.

On the other hand, XGBoost is a scalable tree-boosting system, faster in training, prone to over-fitting, and robust to outliers [28]. The XGBoost model used second-order Taylor expansion and adds a regularization term. It consists of a sequence of regression trees, that iteratively applies gradient descent to boost weak learners (mostly decision trees) to strong learners and improve the performance of the model.

Stacked generalization is an ensemble method that aggregates base models using a model fusion approach. First, multiple heterogeneous base learners are trained with the same data set. Then, the predicted values (outputs) of the base-level learners are used as input features to the higher-level model, the meta-learner, and trained on the validation data set to make the final prediction. In this study, the base learners are ConvLSTM and XGBoost models, while linear regression is set as the meta-learner. (see Fig 2)

1) Bus Running time model: The running time prediction for the next segment can be approached as a time series prediction. The input data for the ConvLSTM model are arranged into a 4-D tensor, $N \times w \times j \times 1$ such that N samples, each with the window of w lagging time-steps and j segment running times. The input data of travel times were detrended using normalization, centring the values with the mean of the segment at the time of day and day of the week.

Besides, this problem can be treated as regression. The structured tabular data with 18 features is fetched to XGBoost Regressor. Its hyper-parameters are tuned by Bayesian optimization. The predicted running times from base models are fed as input to the meta-learner, linear regression.

2) Bus Dwell time model: The dwell time model was developed in congruence with the running time model, except for the dwell time classifier module. In our case, since the GPS sampling frequency is limited to 15 seconds, dwell times less than $15s$ are taken as $brief$ stops. This led to more zeroes in the input data; hence, XGBoost produced negative predictions. To overcome this issue, we first proposed a dwell time classifier module to predict whether the upcoming stop is $brief$. If not, the XGBoost regressor will predict the dwell time. In the case of the ConvLSTM model, we added the *ReLU* activation function at the output layer, to predict non-negative dwell times. The predicted running time and dwell time are merged to get the arrival time to the next stop.

C. Feature important analysis

An explanatory variable analysis can interpret the predictions based on the impacting features for bus travel times. We applied SHapley Additive exPlanations (SHAP) [29]. The contribution of each feature to the model's individual predictions is computed by SHAP based on the ideology of cooperative game theory. It considers all possible coalitions of features and computes the marginal contribution of each feature to the prediction, by comparing the prediction with and without the feature. These values further state that contributions are either positive or negative. Ultimately, the insights from this analysis support public transport authorities in decision-making and regulating buses.

IV. Experiments

The data used in this study was collected from the AVL system fixed on buses in Kandy, Sri Lanka. The raw, crude GPS data were processed to extract the segment running times and dwell times. The detailed explanation and challenges in processing the raw GPS data were presented in a study [2], and the Python-based framework is open-sourced [2]. The data set can be provided upon request. Route No. 654 from Kandy to Digana terminals with 14 bus stops was considered. From the data of nine months in 2022, we used 36 weeks of data for training and 6 for testing. Outliers were treated by capping after clustering bus stops based on demand and segments by length. Given the 15-minute time resolution from 6 a.m. to 7 p.m., we set the fixed window size, $w = 52$, to allow patterns in a day in the training phase.

The proposed model was implemented in Python using the Keras framework and trained using the RMSProp optimization algorithm. The proposed multi-model ensemble approach by stacking was compared against baseline models such as; (1) ConvLSTM segment-based model [24], (2) XGBoost segment-based model [8], (3) Multi-model with ConvLSTM for link travel time prediction and Exponential smoothing (ES) for the dwell time prediction which is proposed by Petersen et al. [11], and (4) Two independent ConvLSTM models for predicting running time and waiting time [13]. In addition, the dwell time *classifier* module is compared among XGBoost and KNN classifiers to choose the best model for that module within the dwell time model.

The performance of the proposed model and baselines is evaluated by; (1) *mean absolute error* (MAE), (2) *root mean square error* (RMSE), and (3) *mean absolute percentage error* (MAPE) as formulated in the eqs (2) to (4), where y_i is the true times and $\hat{y}_i$ is the predicted travel times of N samples.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{2}$$

[2] https://github.com/aaivu/gps2gtfs

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \quad (3)$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \quad (4)$$

V. Results and Analysis

This section presents the results of the above experiments, compares the performance of the proposed model with baseline models, and analyzes the feature importance.

A. Performance analysis of the proposed model

The table II below shows the overall evaluation results of the proposed multi-model ensemble of stacked generalization with other existing approaches. The results show that our proposed multi-model ensemble of ConvLSTM and XGBoost with the segregation of running time and dwell time prediction outperformed well over the other existing models. On average, our model has an 11% reduction in MAE than the four selected approaches.

TABLE II

Comparison of the performance of the proposed model with other existing approaches by using error metrics

Model	MAE (s)	MAPE (%)	RMSE (s)
ConvLSTM segment-based [24]	43.1	20.50	71.4
XGBoost segment-based [8]	41.0	23.02	62.3
ConvLSTM + ES [11]	39.2	19.12	63.5
ConvLSTM multi-model [13]	39.7	20.64	61.7
Proposed Multi-model ensemble	**36.2**	**19.01**	**58.2**

We further investigated the prediction performance of the proposed model. We plot the average of the predicted and actual values for running time at different times of the day as shown in Fig 4. Our model predicts the running time precisely at extreme travel conditions at the morning peak (from 6 A.M to 7 A.M). When observing the pattern of prediction, the multi-model ensemble tried to leverage the positive and negative deviations from the actual values caused by XGBoost and ConvLSTM models respectively.

B. Feature important analysis

The computed mean SHAP values of each feature for the dwell time model are plotted in the horizontal bar chart shown in Fig. 5. The $'device_id$ that indicates the driver's behaviour has the most impact in predicting the next dwell times. The drivers tend to intentionally wait for prolonged or skip some stops. Next, every bus stop has various prominence levels and it impacts dwell times. Further, temporal features like dwell time at the stop in the previous time slot $DT(t-1)$ and the same time slot at the previous week $DT(w-1)$ are other important features. However, at the other extreme, whether it's a weekend or holiday has the least influence on the dwell time. Similarly, when interpreting the feature importance of the running time model, we found that temporal features $RT(t-1)$, $RT(w-1)$ and the length of the segment have a significant contribution to running time prediction.

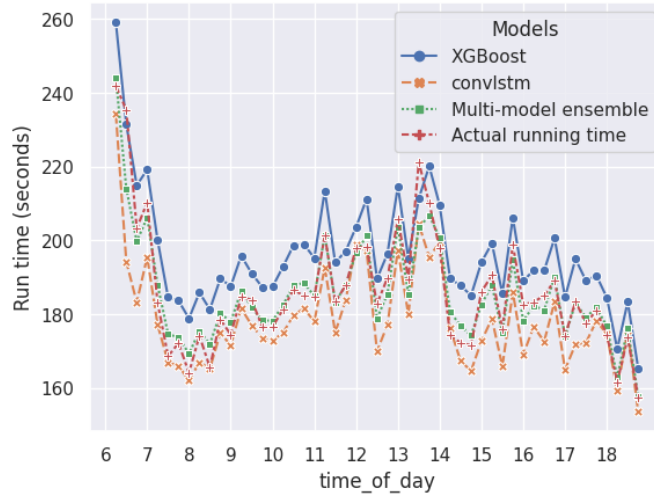

Fig. 4. Variation of the average predicted running times of the proposed model and state-of-the-art models with the actual running times, at different time horizons of a day. Our proposed model (shown in green colour line) can able to predict precisely at extreme conditions

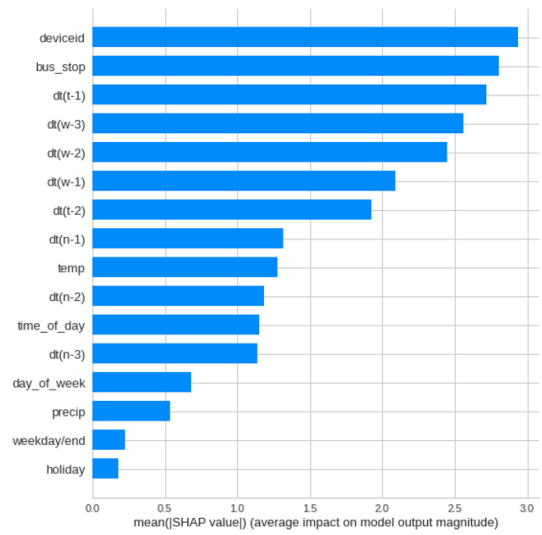

Fig. 5. Horizontal bar plot showing the SHAP values of each feature for dwell time prediction model. It depicts that *device_id* which indicates driver behaviour influences much in predicting dwell time at the next stop

VI. Conclusions and Future works

In this paper, we investigate the limitations of existing methods in predicting bus arrival times in heterogeneous traffic conditions, and the low accuracy of state-of-the-art models to capture the higher variability of travel times. To solve that, a multi-model ensemble approach was used to predict the arrival times under heterogeneous conditions with

the segregation of bus dwell times and running times. For that, we exploited two state-of-the-art models (ConvLSTM & XGBoost) in travel time prediction used in homogeneous traffic. Later, we applied the stacked generalisation ensemble technique with linear regression as the meta-learner to produce the best prediction model for heterogeneous traffic conditions. For the best performance of the XGBoost component, we developed a feature engineering (generation) framework from the available GPS and weather data. Our proposed approach outperformed the state-of-the-art methods on average by 11% in terms of MAE. Further, we have found that driver behaviour, prominence of the bus stop, and antecedent dwell time at the stop have a significant impact on the prediction of arrival times to the next bus stop.

In future, we plan to generalize our proposed model with homogeneous traffic data and develop it as a multi-step ahead prediction to update arrival times at all downstream bus stops in real-time. Further, due to the 'concept drift' in the travel time distribution over time, incremental online learning can be adopted with existing methods. From our insights, a research direction is created to explore the influence of driver behaviour on schedule adherence and bus travel time variations.

ACKNOWLEDGMENT

This research was supported by the Accelerating Higher Education Expansion and Development (AHEAD) Operation of the Ministry of Higher Education, Sri Lanka funded by the World Bank.

REFERENCES

[1] G. Sihag, M. Parida, and P. Kumar, "Travel Time Prediction for Traveler Information System in Heterogeneous Disordered Traffic Conditions Using GPS Trajectories," *Sustainability*, vol. 14, p. 10070, Aug. 2022.

[2] S. Ratneswaran and U. Thayasivam, "Extracting potential travel time information from raw gps data and evaluating the performance of public transit-a case study in kandy, sri lanka," in *2023 3rd International Conference on Intelligent Communication and Computational Techniques (ICCT)*, pp. 1–7, IEEE, 2023.

[3] K. K. Reddy, B. A. Kumar, and L. Vanajakshi, "Bus travel time prediction under high variability conditions," *Current Science*, pp. 700–711, 2016.

[4] R. Padmanaban, L. Vanajakshi, and S. C. Subramanian, "Estimation of bus travel time incorporating dwell time for apts applications," in *2009 IEEE Intelligent vehicles symposium*, pp. 955–959, IEEE, 2009.

[5] A. Abdi and C. Amrit, "A review of travel and arrival-time prediction methods on road networks: classification, challenges and opportunities," *PeerJ Computer Science*, vol. 7, p. e689, Sept. 2021.

[6] S. Nadeeshan and A. S. Perera, "Multi-Step Bidirectional LSTM for Low Frequent Bus Travel Time Prediction," in *2021 Moratuwa Engineering Research Conference (MERCon)*, (Moratuwa, Sri Lanka), pp. 462–467, IEEE, July 2021.

[7] W. Fan and Z. Gurmu, "Dynamic Travel Time Prediction Models for Buses Using Only GPS Data," *International Journal of Transportation Science and Technology*, vol. 4, no. 4, pp. 353–366, 2015.

[8] L. Zhu, S. Shu, and L. Zou, "XGBoost-Based Travel Time Prediction between Bus Stations and Analysis of Influencing Factors," *Wireless Communications and Mobile Computing*, vol. 2022, pp. 1–25, July 2022.

[9] O. Osman, H. Rakha, and A. Mittal, "Application of Long Short Term Memory Networks for Long- and Short-term Bus Travel Time Prediction," preprint, other, Apr. 2021.

[10] Y. Yuan, C. Shao, Z. Cao, Z. He, C. Zhu, Y. Wang, and V. Jang, "Bus Dynamic Travel Time Prediction: Using a Deep Feature Extraction Framework Based on RNN and DNN," *Electronics*, vol. 9, p. 1876, Nov. 2020.

[11] N. C. Petersen, F. Rodrigues, and F. C. Pereira, "Multi-output Deep Learning for Bus Arrival Time Predictions," *Transportation Research Procedia*, vol. 41, pp. 138–145, 2019.

[12] J. Ma, J. Chan, G. Ristanoski, S. Rajasegarar, and C. Leckie, "Bus travel time prediction with real-time traffic information," *Transportation Research Part C: Emerging Technologies*, vol. 105, pp. 536–549, Aug. 2019.

[13] J. Wu, Q. Wu, J. Shen, and C. Cai, "Towards Attention-Based Convolutional Long Short-Term Memory for Travel Time Prediction of Bus Journeys," *Sensors*, vol. 20, p. 3354, June 2020.

[14] N. C. Petersen, F. Rodrigues, and F. C. Pereira, "Multi-output Bus Travel Time Prediction with Convolutional LSTM Neural Network," *Expert Systems with Applications*, vol. 120, pp. 426–435, Apr. 2019. arXiv:1903.02791 [cs, stat].

[15] D. H. Wolpert, "Stacked generalization," *Neural networks*, vol. 5, no. 2, pp. 241–259, 1992.

[16] L. Vanajakshi, S. C. Subramanian, and R. Sivanandan, "Travel time prediction under heterogeneous traffic conditions using global positioning system data from buses," *IET intelligent transport systems*, vol. 3, no. 1, pp. 1–9, 2009.

[17] B. A. Kumar, L. Vanajakshi, and S. C. Subramanian, "Bus travel time prediction using a time-space discretization approach," *Transportation Research Part C: Emerging Technologies*, vol. 79, pp. 308–332, 2017.

[18] B. A. Kumar, A. Achar, D. Bharathi, and L. Vanjakshi, "A Seasonal Modelling Approach Capturing Spatio-Temporal Correlations for Dynamic Bus Travel Time Prediction," in *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*, (Auckland, New Zealand), pp. 503–508, IEEE, Oct. 2019.

[19] A. Achar, D. Bharathi, B. A. Kumar, and L. Vanajakshi, "Bus Arrival Time Prediction: A Spatial Kalman Filter Approach," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, pp. 1298–1307, Mar. 2020.

[20] D. Sun, H. Luo, L. Fu, W. Liu, X. Liao, and M. Zhao, "Predicting Bus Arrival Time on the Basis of Global Positioning System Data," *Transportation Research Record: Journal of the Transportation Research Board*, vol. 2034, pp. 62–72, Jan. 2007.

[21] B. Yu, W. H. Lam, and M. L. Tam, "Bus arrival time prediction at bus stop with multiple routes," *Transportation Research Part C: Emerging Technologies*, vol. 19, pp. 1157–1170, Dec. 2011.

[22] T. Liu, J. Ma, W. Guan, Y. Song, and H. Niu, "Bus Arrival Time Prediction Based on the k-Nearest Neighbor Method," in *2012 Fifth International Joint Conference on Computational Sciences and Optimization*, (Harbin, Heilongjiang, China), pp. 480–483, IEEE, June 2012.

[23] P. He, G. Jiang, S.-K. Lam, and D. Tang, "Travel-Time Prediction of Bus Journey With Multiple Bus Trips," *IEEE Transactions on Intelligent Transportation Systems*, vol. 20, pp. 4192–4205, Nov. 2019.

[24] Z.-Y. Xie, Y.-R. He, C.-C. Chen, Q.-Q. Li, and C.-C. Wu, "Multistep Prediction of Bus Arrival Time with the Recurrent Neural Network," *Mathematical Problems in Engineering*, vol. 2021, pp. 1–14, Mar. 2021.

[25] B. Yu, H. Wang, W. Shan, and B. Yao, "Prediction of Bus Travel Time Using Random Forests Based on Near Neighbors: Prediction of bus travel time," *Computer-Aided Civil and Infrastructure Engineering*, vol. 33, pp. 333–350, Apr. 2018.

[26] B. Qiu and W. D. Fan, "Machine Learning Based Short-Term Travel Time Prediction: Numerical Results and Comparative Analyses," *Sustainability*, vol. 13, p. 7454, July 2021.

[27] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-c. Woo, "Convolutional lstm network: A machine learning approach for precipitation nowcasting," *Advances in neural information processing systems*, vol. 28, 2015.

[28] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," in *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp. 785–794, 2016.

[29] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions,"